FactGrid:Cuneiform Project
Factgrid

Wikidata:Lexicographical data
Wikidata

FactGrid Cuneiform 2024

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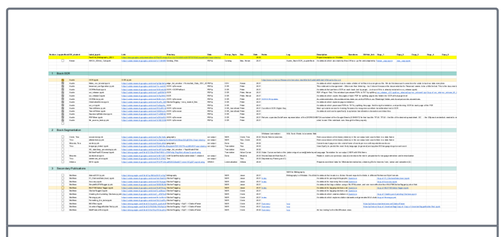


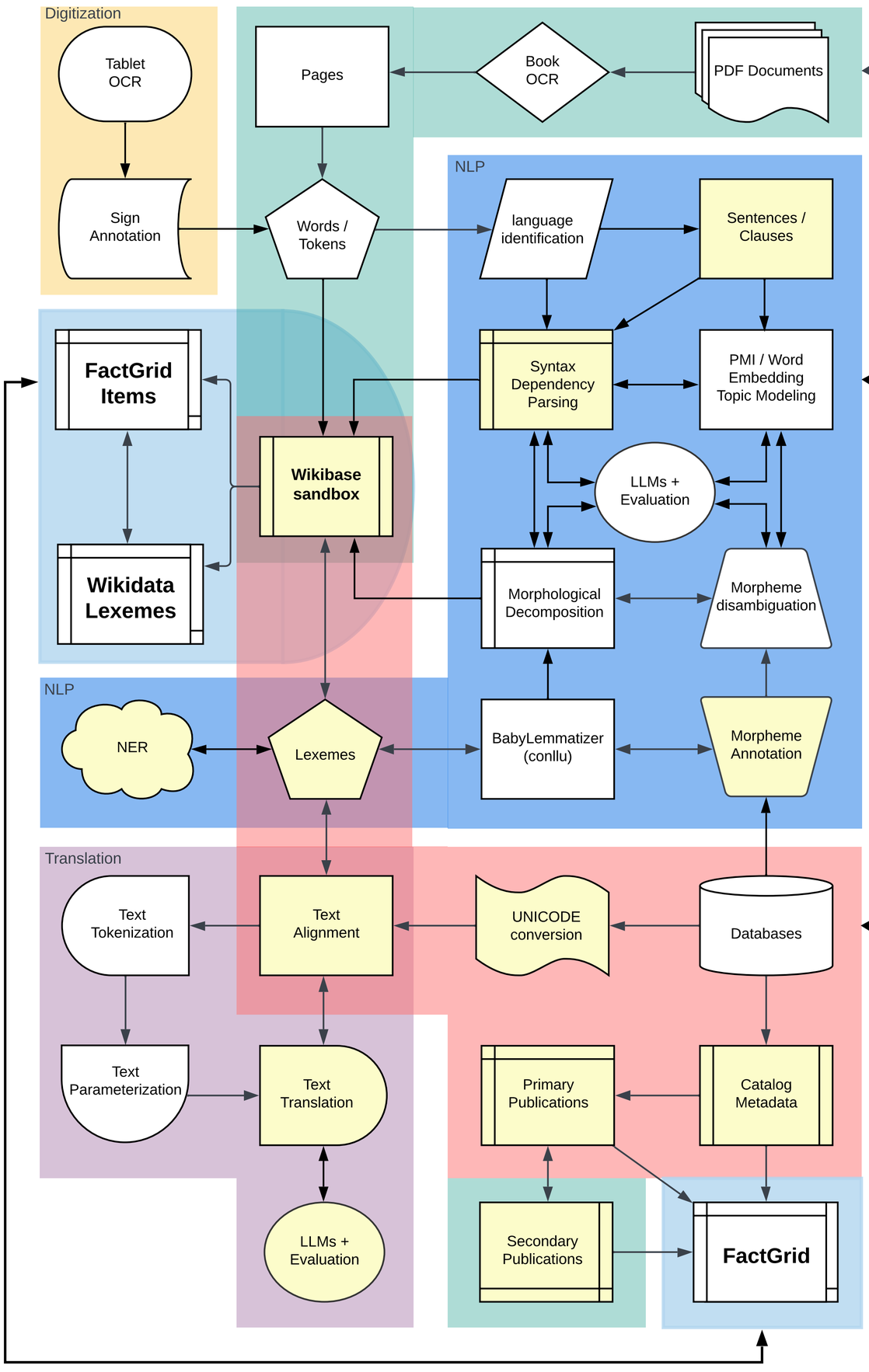
FactGrid_Cuneiform_AWCA_Notebooks
Google Sheets

Jupyter Notebooks for Knowledge Graphs & LLMs (LLaMA & GPTs)

Project Description: This semester we are working with Large Language Models (LLM) of different types (e.g. GPTs, Llama, mBERT, etc.). We have written a number of prompts to run different experiments on the cuneiform data, to see if the LLMs can out-perform the existing NN-approaches we have built in previous semesters. We have the ability to run these experiments in Jupyter Notebooks, in order to ensure the reproducibility of each stage of the research. We have divided these experiments into 4 groups: (1) classification (2) prediction (3) disambiguation (4) recognition. Each group will work on a series of research questions having to do with solving different data challenges in working with 400k texts in Sumerian, Akkadian, Hittite, and other low-resource languages from antiquity. The goal of this collaborative process is to demonstrate the current capability of the LLMs for working with the ancient cuneiform texts in different ways. We plan to publish the results of the research for the semester in FactGrid, Wikidata, and in GitHub with Jupyter Notebooks. Each student that participates will receive detailed authorship attribution.

Pipeline Description: The diagram below shows how each notebook fits together in a series of tasks in terms of digitization, data acquisition, data formatting, annotating, translating, and ultimately linking with Wikidata lexemes and FactGrid items. The boxes highlighted in yellow were the tasks which we developed further this semester.
Project Goal: To develop the optimal workflow for LOD lemmatization and publication in FactGrid. Once the workflow is evaluated and optimized, it can then be implemented in a chain-of-thought using LLMs as **Multi-Agent Frameworks**, which are "agentic" workflows that spawn many LLM agents to execute tasks like an informed swarm of bees. By setting up the tasks in advance, we can constrain the inference each agent needs to execute a task, and ensure the workflow is fully replicable for the entire scale of the project, similar to **RISC**: reduce instruction set computing.
Nest Steps: Next semester we will operationalize the publication of our results in the Wikidata and FactGrid Knowledge Graphs, using Jupyter Notebooks and APIs.

Linked Open Data (LOD) Pipeline



Descriptions for LOD Pipeline

Books & PDF Data

More than 200,000 books, journals and articles in 20+ languages. Subject matter is ancient languages, cultures, history before 500 B.C.

Goals for data set:
- Lemmatize data (to resolve OCR)
- Identify cuneiform (Sumerian, Akkadian, Hittite, etc.)
- Add new / unique words to Wikidata
- Distant reading analysis (with PMI / embedding)
- Language model evaluation

NLP

Natural Language Processing using SpaCy and rule-based dictionary methods.

Morphology:
- Part-of-Speech tags (POS) with SpaCy / UD
- BabyLemmatizer (conllu)

Syntax:
- Dependency tags with Spacy / UD
- Development for BabyLemmatizer (conllu)

NER:
- Proper nouns in Oracc have unique tags, which we extend to include named entites in all languages
- Economic classification: area, weight, capacity (volume), weight, and simple counts (int.).

Databases & Web Data

More than 400,000 catalog entries for primary source texts in cuneiform. Databases include: CDLI, ORACC (ePSD, BDTNS, ETCSRI, etc.), OARE.
Web data include: KeiBi, SumZett, Zotero, Zenodo, Sumerian / Babylonian Verb Conjugator, AICC, etc.

Goals for data set:
- Lemmatize data (BabyLemmatizer vs. LLaMA)
- Convert to cuneiform (transliteration to Unicode cuneiform signs)
- Add new / unique words to Wikidata
- Build language modes for Sumerian and Akkadian

Machine Translation (MT)

Tools for Sumerian and Akkadian translation are in development and in need of evaluation. We have a series of notebooks for evaluating cuneiform translation, based on the Akkademia GitHub repository.

Goals for translation:
- Use Lemmatized data for evaluation
- Gather more gold standard translations
- Integrate LLM architectures
- Compare architectures with gold standard
- Build Language models for Sumerian and Akkadian

2024 Data Science Discovery Team

Full Name	Project roles	Graduation
Athena He	NLP / Alignment	Graduated
Austin Pereira	OCR	2025
Christian Karren	MT	2025
Daisy Wang	Citation Analysis	2025
Diego Sotomayor	NER	2024
Iris Tsui	JupyterBook	2025
Ishan Monie	LLM / Citation	2025
Kenan Jiang	NLP / Dep.	Graduated
Lirui Huang	NLP / Alignment	2026
Melinee Her	Databases	2025
Micaela Montes	NLP / Alignment	2025
Nasim Borazjanizadeh	LLM / NLP / MT	2024
Olivia McCauley	LLM / MT	MIDS Student
Samuel Goesling	Wikidata	2024
Tina Chen	NLP / Dep.	2024
Win Moe	NER	2026

Wikidata & FactGrid

Wikidata contains linguistic annotations (LLOD) for more than 1,187 different languages, making this the richest resource for languages in a knowledge graph. We leverage these tools in order to lemmatize our data and build up the existing languages from our dataset.

Wikidata Lexicography: we are currently building **Sumerian** and **Akkadian** lexemes, forms, senses in Wikidata.

Goals for Wikidata set:
- Add all lexemes from major lexicons and dictionaries
- Add all senses linked to Wikidata Q-items
- Add all forms linked to primary source IDs in FactGrid
- Add grammatical units, especially morphology
- Add syntax dependencies (usage examples)
- Machine Translation Sumerian and Akkadian

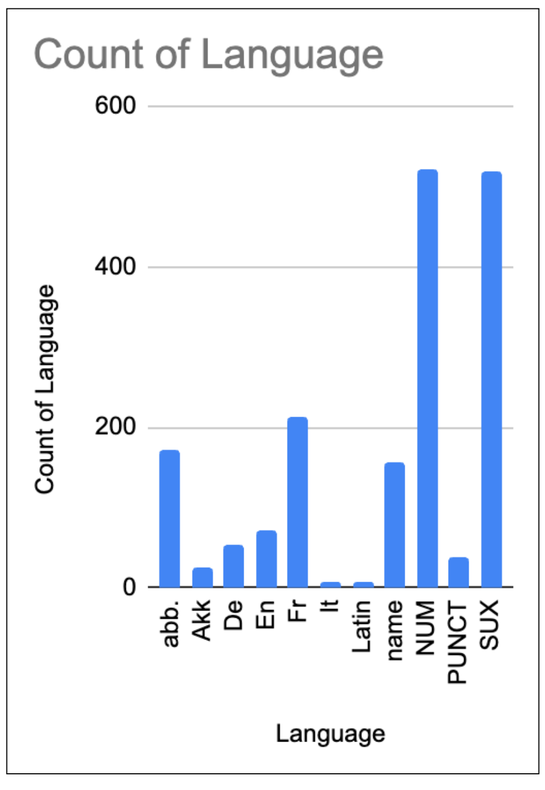
FactGrid Publication: each formatted version of every document can be published in FactGrid transcripts (or wiki pages), including the catalog metadata as triple statements and the different formats of the text (i.e. transliterations, lemmatizations, and translations). We have also developed an RDFa format using HTML which will provide hyperlinked text for users to have instant access to the LOD Wikidata lexemes.

Implementation and the Current Results

LOD Publication: Through the implementation of the LOD pipeline described above, we have an automated method for the digitization of cultural heritage in the form of book-PDFs and images. By utilizing Wikidata lexemes we are able to address the issues raised in the post-processing of OCR errors, which simultaneously contributes high-value linguistic publications to the rich language resources in current development through the Wikidata lexicography portal. The resulting resource is a knowledge graph capable of training LLMs for multi-lingual annotation and translation tasks, including currently unsupported languages like Sumerian. In applying this workflow this semester, we found the following descriptive logic is capable of yielding the described outcomes: for a given document we have n-tokens lemmatized with Wikidata L-ids. The result will be one of two outcomes, both resulting in LOD publications in Wikidata Lexecography:

A. If the lemmatized tokens represent all of the document's tokens, then we can publish the text with hyperlinks for each lemma, e.g. the transcript of the document with L-id links to the Wikidata lexicon (and provide links to each form and/or sense for hyperlinked text in FactGrid). Note, open wikibase publication will be available only for those documents which are out of copyright for public / fair use.

B. Otherwise, we can publish the remaining unmatched tokens as Wikidata lexemes or forms, and then lemmatize the document, with the same outcomes as in A. This will further develop the 1180+ languages in Wikidata, as well as enhance the accuracy of any LLM which uses this data.

Text	Lemma	POS	Dep	Wikidata
Ces	ce	DET	det	http://www.wikidata.org/entity/L9134
pages	page	NOUN	nsubj:pass	http://www.wikidata.org/entity/L34126
sont	être	AUX	aux:pass	http://www.wikidata.org/entity/L9944
consacrées	consacrer	VERB	ROOT	http://www.wikidata.org/entity/L16495
à	à	ADP	case	http://www.wikidata.org/entity/L19240
une	un	DET	det	http://www.wikidata.org/entity/L13103
étude	étude	NOUN	obl:arg	http://www.wikidata.org/entity/L17969
de	de	ADP	case	http://www.wikidata.org/entity/L2379
ak	ak	ADP	det	http://www.wikidata.org/entity/L726793

Wikibase Lemmatization Example